

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure for Determination of Water Insulating Liquid by Coulometric Karl Fischer Titration SI Analytics TitroLine 7500

Equipment Number: EQ-69 & EQ-70
Serial Number: 10047950 & 10048514



1. PURPOSE

This standard operating procedure is intended to provide users with a guide on the operation and maintenance of the Karl Fischer Coulometer.

2. SCOPE

This standard operating procedure is employed in the GLTS primarily for measuring the range of water content of insulating oil in automatic conditioning, drift determination and real time curve titration and results calculation output by double platinum electrode Karl Fischer coulometric cell.

3. DISTRIBUTION

This operational procedure is distributed to the following:

<u>Copy No.</u>	<u>Copyholder</u>
Original	Quality Manager
2	Technical Manager and Chemists

4. TERMS AND DEFINITIONS

- 4.1 Drift: Background moisture.
- 4.2 Hydranal: Karl Fischer reagent.
- 4.3 Generator electrode: Electrode that delivers an electrical current inside the vessel.
- 4.4 Detector electrode: Electrode that measures the change in current.

5. RESPONSIBILITIES

- 5.1 Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

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6. SAFETY PRECAUTIONS

- 6.1 Use a proper electrical outlet of correct voltage and current handling capacity.
- 6.2 Disconnect from the power supply prior to maintenance and servicing.
- 6.3 Ensure that the operating switch operates properly.

7. PRINCIPLES OF OPERATION

7.1 Application

7.1.1 The coulometric Karl Fischer titration is a version of the classical water determination method developed by Karl Fischer. The traditional method utilizes a methanolic solution of iodine, sulphur dioxide and a base as buffer (Hydranal). Several reactions run in the titration of a water-containing sample and can be summarized by the following overall equation:



7.1.2 According to the above equation, I_2 reacts quantitatively with H_2O . This chemical relation forms the basis of the water determination.

7.1.3 In the coulometric Karl Fischer titration, the iodine needed is generated directly in the electrolyte by electrochemical means ("electronic buret"). The rigorous quantitative relationship between the electric charge and the amount of iodine generated is used for high-precision dispensing of the iodine.

7.1.4 As the coulometric Karl Fischer method is an absolute determination no titer need be determined. It is necessary only to ensure that the reaction which generates the iodine runs with 100% current efficiency. With the reagents available today this is always the case.

7.1.5 The end point is indicated voltametrically by applying an alternating current of constant strength to a double Pt electrode. This results in a voltage difference between the Pt wires of the indicator electrode which is drastically lowered in the presence of minimal quantities of free iodine. This fact is used to determine the end point of the titration.

8. OPERATIONAL DETAILS

8.1 Operational Procedure

- 8.1.1 The titration vessel has been prepared and the Coulometer is switched on. In the display:
- 8.1.2 Wait until the drift is under 20 mg/g and stable.

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- 8.1.3 Press <START> and inject the first sample.
- 8.1.4 Enter the weight (sample) and confirm it with <ENTER>.
- 8.1.5 After the titration the result is displayed and printed out by the internal printer.
- 8.1.6 The titration vessel is continuously kept dry and the current drift is displayed.
- 8.1.7 If you want to determine further samples press <START> again and inject the next sample.

9. Maintenance

9.1 Titration Vessel Setup

- 9.1.1 Attach titration vessel with holder to the support rod and place stirring bar in titration vessel.
- 9.1.2 Cut ground joint sleeves to the correct lengths and use them for all the joints of the inserts.
- 9.1.3 Insert detector electrode in the Right-hand joint opening, screw on electrode cable and plug it into the "DE" socket of the Coulometer. Mark the screw head of the electrode cable so that it is impossible to confuse the indicator and generator electrodes!
- 9.1.4 Insert generator electrode in the Left-hand joint opening, screw on electrode cable and plug it into the "GE" socket of the Coulometer.
- 9.1.5 Fill the drying tube with silica gel and insert it into generator electrode. Place septum in the screw cap and screw this onto the titration vessel. Only tighten it enough to ensure that it is tight. (The septum should not be deformed)
- 9.1.6 Fill titration vessel with 80-100 ml Hydranal reagent.
- 9.1.7 Close last joint opening: either with glass stopper, aspiration device or gas inlet from oven.
- 9.1.8 Detector Electrode: A new detector electrode may require a certain running-in period for the formation of the surface. This may cause unusually long titration times and measurement results which are too high. These phenomena vanish after a short period of use. In order to speed up the running- in of a new indicator electrode the Coulometer can be conditioned overnight, for example. A polluted detector electrode can be carefully cleaned with an abrasive cleansing agent (aluminum oxide or toothpaste). After

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cleaning it should be rinsed with methanol. The two Pt wires of the indicator electrode should be as parallel to one another as is possible.

9.1.9 Cleaning: The electrolyte solution can normally be exchanged without any special cleaning of the parts being necessary. If cleaning is necessary, then care should be taken that the Pt grid of the generator electrode is not damaged.

9.1.10 Septum should be changed frequently if it is damaged.

9.1.11 Molecular Sieve can be changed or restored by drying it in the oven at 150 °C.

REFERENCES

- [1] ASTM D1533 Standard test method for water in insulating liquids by coulometric Karr Fischer titration.
- [2] IEC 60814 insulating liquids - Oil-impregnated paper and pressboard - Determination of water by automatic coulometric Karl Fischer titration
- [3] SI Analytics TitroLine 7500 KF Coulometer User's Manual.

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